

JEE MAIN REVISION MASTER: ELLIPSE SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 116 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Eccentricity, Foci & Tangents of Ellipse	116 PYQs	Ellipse
Moderate	Focal Properties & Auxiliary Circle	3 PYQs	Focal Geometry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Eccentricity, Foci & Tangents of Ellipse	$e = \sqrt{1 - \frac{b^2}{a^2}}, \quad \text{Foci: } (\pm ae, 0), \quad \text{LR: } \frac{b^2}{a}$ $\text{Tangent: } y = mx \pm \sqrt{a^2 m^2 + b^2}, \quad \text{Director: } x^2 + y^2 = a^2 + b^2$
Focal Properties & Auxiliary Circle	$SP + S'P = 2a = \text{Major Axis}, \quad \text{Auxiliary Circle: } x^2 + y^2 = a^2,$ $\text{Parametric: } (a \cos \theta, b \sin \theta)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Eccentricity, Foci & Tangents of Ellipse (Very High Yield — 116 PYQs)

Core Principle: Standard ellipse equation, focal distance sum property $SP + S'P = 2a$, eccentricity $e < 1$, and director circle.

■ **Caution / JEE Pitfall:** If $b > a$ (vertical ellipse), major axis is along y-axis and $e = \sqrt{1 - a^2/b^2}$ with foci at $(0, \pm be)$.

■ **Speed Trick:** Product of perpendiculars from the two foci onto any tangent to the ellipse is always constant and equal to b^2 .

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\frac{x^2}{f(a^2+7a+3)} + \frac{y^2}{f(3a+15)} = 1$ represent an ellipse with major axis along y-axis, where f is a strictly dec..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a < b$, pass through the point (4, 3) and have eccentricity $\frac{\sqrt{5}}{3}$..." [Ans: D]
- [2026 Evening Shift] Q#1 (MCQ): "If the points of intersection of the ellipses $x^2 + 2y^2 - 6x - 12y + 23 = 0$ and $4x^2 + 2y^2 - 20x - 12y + 35 = 0$ lie on a circle of radius r and ..." [Ans: N/A]

3.2 Focal Properties & Auxiliary Circle (Moderate Yield — 3 PYQs)

Core Principle: Focal radius property $SP + S'P = 2a$, auxiliary circle projections, and parametric coordinate representation.

■ **Caution / JEE Pitfall:** In an ellipse, the eccentric angle η is measured on the auxiliary circle, NOT from the center to the point on the ellipse.

■ **Speed Trick:** Maximum area of a rectangle inscribed in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $2ab$.

Representative Source PYQs:

- **[2025 Evening Shift] Q#1 (MCQ):** "Let the ellipse $3x^2 + py^2 = 4$ pass through the centre C of the circle $x^2 + y^2 - 2x - 4y - 11 = 0$ of radius r. Let f_1, f_2 be th..." **[Ans: N/A]**
- **[2024 Morning Shift] Q#3 (MCQ):** "Let the line $2x + 3y - k = 0, k > 0$, intersect the x-axis and y-axis at the points A and B, respectively. If the equation of the circle..." **[Ans: N/A]**
- **[2015 Shift] Q#1 (MCQ):** "The area (in sq. units) of the quadrilateral formed by the tangents at the end points of the latera recta to the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

Let an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a < b$, pass through the point $(4, 3)$ and have eccentricity $\frac{\sqrt{5}}{3}$. Then the length of its latus rectum is :

- (A) $\frac{4\sqrt{5}}{3}$
- (B) $2\sqrt{5}$
- (C) $\frac{7\sqrt{5}}{3}$
- (D) $\frac{8\sqrt{5}}{3}$ **[Correct]**

Correct Answer: D

Solution Analysis: Since $a < b$, the major axis is along the y -axis. So for the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, we use the eccentricity formula $e = \sqrt{1 - \frac{a^2}{b^2}}$. Given $e = \frac{\sqrt{5}}{3}$, so $e^2 = \frac{5}{9}$. Hence, $1 - \frac{a^2}{b^2} = \frac{5}{9}$ which gives $\frac{a^2}{b^2} = 1 - \frac{5}{9} = \frac{4}{9}$. Therefore, $a^2 = \frac{4}{9}b^2$. Now the ellipse passes through $(4, 3)$, so it satisfies the equation: $\frac{4^2}{a^2} + \frac{3^2}{b^2} = 1$. That is, $\frac{16a^2}{b^2} + \frac{9b^2}{b^2} = 1$. Substitute $a^2 = \frac{4}{9}b^2$: $\frac{16}{\frac{4}{9}} + \frac{9}{1} = 9$